describe strains of bacteria that are resistant to the majority of antibiotics commonly used today.

9.2.2 Fermented beverages

Microbes especially yeast is being used from time immemorial for the production of beverages like wine, beer, whisky, brandy and rum. Wine is among the oldest alcoholic beverages known and is produced by fermentation of fruit juice by yeast. **Zymology** is an applied science which deals with the biochemical process of fermentation and its practical uses.

Saccharomyces cerevisiae commonly called brewer's yeast is used for fermenting malted cereals and fruit juices to produce various alcoholic beverages. Wine and beer are produced without distillation, whereas whisky, brandy and rum are obtained by fermentation and distillation.



The Pasteur effect is the inhibiting effect of oxygen on the fermentation process.

Oenology is the science and study of **wine** and wine making. Wine is made from the fermentation of grape juice. Grape juice is fermented by various strains of *Saccharomyces cerevisiae* into alcohol. Grape wine is of two types, red wine and white wine. For red wine, black grapes are used including skins and sometimes the stems also are used. In contrast white wine is produced only from the juice of either white or red grapes without their skin and stems.

Beer is produced from germinated barley malt grain by *Saccharomyces carlsbergensis* or *Saccharomyces cerevisiae*. **Rum** is made from fermented sugarcane or molasses or directly from sugarcane juice by *Saccharomyces cerevisiae*. **Whisky** is a type of distilled alcoholic beverage made from fermented grain mash by *Saccharomyces cerevisiae*.

Alcohol content in various beverages

- Beer contains 3 to 5 percent of alcohol.
- Wine contains 9 to 14 percent alcohol. Wine coolers are made of wine mixed with carbonated water and flavourings. Wine coolers have about 4 to 6 percent alcohol.
- Distilled spirits such as whiskey, gin, scotch and vodka usually contain 35 to 50 percent alcohol.

In some parts of South India, a traditional drink called **pathaneer** is obtained from fermenting sap of palms and coconut trees. A common source is tapping of unopened spadices of coconut. It is a refreshing drink, which on boiling produces jaggery or palm sugar. When pathaneer is left undisturbed for few hours it gets fermented to form **toddy** with the help of naturally occurring yeast, to form a beverage that contains 4 percent alcohol. After 24 hours **toddy** becomes unpalatable and is used for the production of vinegar.

Saccharomyces cerevisiae is the major producer of ethanol (C_2H_5OH). It is used for industrial, laboratory and fuel purposes. So ethanol is referred to as **industrial alcohol**. Bacteria such as *Zymomonas mobilis* and *Sarcina ventriculi* are also involved in ethanol production. The principal substrates for the commercial production of industrial alcohol include molasses or corn, potatoes and wood wastes. The process of ethanol production starts by milling a feed stock followed by the addition of dilute or fungal **amylase (enzyme)** from *Aspergillus* to break down the starch into fermentable sugars. Yeast is then added to convert the sugars to ethanol which is then distilled off to obtain ethanol which is upto 96 percent in concentration. The two most common type of biofuels in use today are ethanol and biodiesel, both of them represent the first generation of biofuel technology. Ethanol is often used as a fuel, mainly as a biofuel additive for gasoline.

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